How Vague Meanings Shape Interpretation: Evidence from Approximators

Word Count: 3,904

Abstract

A recent Bayesian account of the interpretation of numerical approximation expressions (Égré et al., 2023) makes the central prediction that regardless of the prior used, the posterior distribution of possible numbers induced by an expression of the form *around n* should be more peaked toward its central value *n* than the posterior induced by an equivalent expression of the form *between m and m′*. In this paper, we test and verify this prediction using a novel online probability-elicitation paradigm. Our results confirm the claim that semantically vague expressions like *around n* convey distinctive probabilistic information regarding what the distribution of possible numbers might be, in a way that uncertain but semantically precise expressions like *between m and m′* do not.

Keywords: Vagueness ; Approximation ; Probability ; Around ; Between ; Bayesianism

1 Two kinds of approximators

In recent years, the interpretation of vague expressions has been argued to support a Bayesian approach to linguistic communication (Frazee & Beaver, 2010; Lassiter & Goodman, 2017; Sutton, 2018; Égré et al., 2023). Several computational models of the interpretation of vague expressions have been proposed, in particular using the Rational Speech Act framework (Goodman & Stuhlmüller, 2013). In this paper, we report on several experiments which test a new and distinctive prediction of these models, focusing on the interpretation of so-called *approximators* (Sauerland & Stateva, 2007). Approximators are modifiers such as *around*, *roughly*, *approximately*. They constitute a subset of what Lakoff (1973) called "hedges" ("words whose job is to make things fuzzier or less fuzzy"). When applied to numerals, approximators end up denoting a wider range of possible values than the numeral they modify.

For example, (1a) could be uttered if 17, 20, or 26 people came to the party. More specifically, *around n* seems to denote an implicit interval which contains the numeral n itself, and which is further constrained by contextual and pragmatic considerations (Channell, 1994; Lasersohn, 1999; Krifka, 2007; Solt, 2015). Thus, Solt (2015) proposes that approximated numerals such as *around n* denote an interval centered on n, whose size is half of the granularity G of a contextually given scale S_G in which n is a mark (e.g., if $G = 10$, and $S_G = [0, 10, 20, 30, ...]$, *around 20* selects the interval $[15, 25]$).

Lower-and-upper-bounded numerical expressions, such as *between 15 and 25* in (1b), also denote an interval. The bounds of this interval, unlike the ones of an interval evoked by *around n*, are precise and context-independent. (1b) will not be felicitous if 14 or 26 people came to the party, but will be if 15 or 25 people came. In other words, *between 15 and 25* explicitly denotes the precise interval $[15, 25]$.

(1) *How many people came to the party?*
   a. **Around** 20 people came to the party.
   b. **Between** 15 and 25 people came to the party.

Hence, even if both approximated numerals (like *around 20*) and lower-and-upper-bounded numerals (like *between 15 and 25*) express uncertainty about an exact numerical value k, they are not quite synonymous. *between m and m′* conveys uncertainty within a fully specified range (namely, the interval $[m, m']$), whereas *around n* conveys uncertainty within an underspecified range. This empirical landscape gives rise to a twofold question. What is the exact nature of the underspecification characterizing the semantics of *around*? And why is *around n* used at all, if, for each value of n, it is possible to find an expression of the form *between m and m′* that precisely and overtly denotes the intended interval?

A solution to the above questions, which also does more justice to the inherent vagueness conveyed by *around*, stems from two observations. The first is that *around n* is sorites-susceptible (Smith, 2008; Lassiter & Goodman, 2017; Égré et al., 2019): intuitively, if k can be taken to be *around n*, then so do $k-1$ and $k+1$ (if not to the same exact degree as k, at least to a close degree). This is unlike *between*, which does not have this property: while 10 is determinately between 10 and 20, 9 is determinately not. This cannot be captured if we assume that *around n* denotes one sharp interval, even if it is taken to be context-dependent: once the size of its interval is set by the context, *around n* gets assigned a *between*-like interpretation.

The second observation is that *around n* conveys that numerical values closer to n are more likely than values more remote from n, and that *between m and m′* does not produce the same effect (Égré, 2022; Égré et al., 2023): These two observations motivate a probabilistic account of approximated and lower-and-upper bounded numerals, which associates *around n* and *between m and m′* with distinct probability distributions on numerical values.

Égré et al. (2023) provide such an account within a Bayesian framework, by stating distinct truth conditions for *around* and for *between*, leading to distinct probabilistic update rules for both expressions. Their approach formally accounts for the previous two observations. In particular, it makes a prediction that can be informally stated as a "peakedness" contrast. Consider two expressions of the form *x is around* n and *x is between m and m′*, respectively yielding the posteriors $\mathbb{P}_a$ and $\mathbb{P}_b$, and assume $\mathbb{P}_a$ and $\mathbb{P}_b$ share the same support (i.e. set of values having a positive probability). Then, regardless of what the prior distribution over possible values of x was, $\mathbb{P}_a$ will be more peaked toward its central value than $\mathbb{P}_b$.

This result is used by Égré et al., 2023 to argue that *around* can be more informative in some contexts than *between* to help the listener figure out the speaker's observation. However, it has not been confronted to actual data. In this paper, we propose to test the empirical validity of this prediction via a series of online posterior-elicitation experiments, in order to establish whether *around* is more than a shorthand way of expressing a fixed implicit *between*, and to validate the insights that these expressions communicate uncertainty in different ways. In Section 2, we present Égré et al.'s account and its predictions. Section 3 presents the design of the experiments, and Section 4 presents the results. Section 5 presents some additional analyses of the participants' data, and Section 6 draws broader lessons from the account.

2 A Bayesian account of *around* and *between* and its predictions

In the spirit of previous probabilistic accounts of pragmatic inference (Lassiter & Goodman, 2013; Qing & Franke, 2015; Bergen et al., 2016), Égré et al., 2023 assume that when a listener with a prior distribution about some random variable x (say the number of people at a party) receives as true a relevant utterance u (such as *x is around n* or *x is between m and m′*), they update their prior on x by conditioning on u being true. When u is of type "*x is between m and m′*", the inference is only about x. When u is of type "*x is around n*", the inference is both about the value x of interest and about the half-width of the interval selected by "around", which is also a random variable.

2.1 *Between*

We take a message $u =$ "*x is between m and m′*", where m and m' are fixed, to be true iff x is in the interval $[m, m']$. The update rule in (2) follows straightforwardly. This update crucially keeps the *relative* probabilities of the values within the interval as they were according to the prior.[1]

(2) $$\mathbb{P}[x = k \mid x \text{ is between } m \text{ and } m'] \propto \begin{cases} \mathbb{P}[x = k] & \text{if} \quad k \in [m, m'] \\ 0 & \text{if} \quad k \notin [m, m'] \end{cases}$$

[1]The notation $\propto$ indicates equality modulo multiplication by a constant factor ensuring that the posterior distribution sums up to 1.

2.2 *Around*

Égré et al. posit that the meaning of *around* is as in (3), where t is taken to be a free, *underspecified* variable (unlike Solt's 2015 granularity parameter which is context-dependent but *determinate* in any given context).

(3) A quantity x is around n iff $x \in [n-t, n+t]$

The variables x and t are viewed as random variables and the listener starts with a prior joint distribution over (x, t) which makes them independent:

(4) $\mathbb{P}[x = k, t = i] = \mathbb{P}[k] \times \mathbb{P}[i]$

Under uncertainty about both x and t, a listener can reason that the sentence is more likely to be true if x is closer to n, since the closer it is to x, the more likely it is that t is large enough to ensure that $x \in [n-t, n+t]$.

The posterior induced by *around n* can be computed as follows:

(5)

$$\begin{aligned}
\mathbb{P}(x = k \mid \textit{around } n) &= \sum_i \mathbb{P}(x = k,\, t = i \mid x \in [n-t,\, n+t]) \\
&= \sum_i \frac{\mathbb{P}(x = k,\, t = i,\, x \in [n-t,\, n+t])}{\mathbb{P}(x \in [n-t,\, n+t])} \\
&= \sum_i \frac{\mathbb{P}(x = k,\, t = i,\, k \in [n-i,\, n+i])}{\mathbb{P}(x \in [n-t,\, n+t])} \\
&\propto \sum_i \mathbb{P}(x = k)\, \mathbb{P}(t = i)\, \mathbb{1}_{k \in [n-i,\, n+i]} \\
&\propto \mathbb{P}(x = k) \sum_{i=|n-k|}^{n} \mathbb{P}(t = i)
\end{aligned}$$

The presence in (5) of a sum whose total number of terms $n - |n-k| + 1$ *decreases* as the distance $|n-k|$ between n and k *increases*, translates into the following fact: the closer k is to n, the higher the posterior assigned to k given *u* = *around n*. This itself stems from the observation that a value of k close to n belongs to *more* intervals centered around n, than a value of k' remote from n does (as schematized in Figure 1).

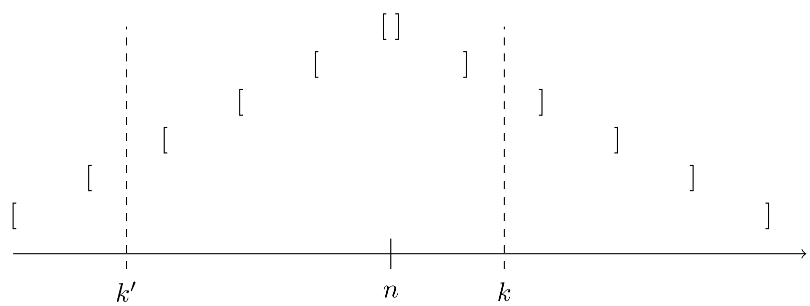


Figure 1: k (closer to n) belongs to four intervals centered around n, while k' (further from n) only belongs to two such intervals.

2.3 Key prediction: the Ratio Inequality

A direct consequence of the Bayesian updates related to *around n* and *between m and m′* is the following. Consider two possible numerical values k and k', such that k is closer to n than k' is, and two posterior distributions $\mathbb{P}[x = k \mid$ *x is around n*$]$ and $\mathbb{P}[x = k \mid$ x is *between m and m′*$]$ sharing the same support S (with $k, k' \in S$). Given this, the ratio of the posterior probabilities of k and k' is predicted to be greater for $\mathbb{P}[x = k \mid$ *x is around n*$]$ than for $\mathbb{P}[x = k \mid$ *x is between m and m′*$]$.

In other words, updating the prior with $u = $ *around n* leads to a greater ratio of posteriors, than updating the same prior with $u = $ *between m and m′*. This is formalized in (6), which Égré et al. call the Ratio Inequality. (6) does not depend on the prior distribution over k. It is therefore possible to test it on multiple listeners, who by default may have distinct priors.

(6) $$\frac{\mathbb{P}[k \mid \textit{around n}]}{\mathbb{P}[k' \mid \textit{around n}]} \geq \frac{\mathbb{P}[k \mid \textit{between m and m}']}{\mathbb{P}[k' \mid \textit{between m and m}']}$$

But there is a way to test (6)'s empirical validity in a more robust fashion. Given two distributions $\mathbb{P}[k \mid \textit{around n}]$ and $\mathbb{P}[k \mid \textit{between m and m}']$ sharing the same support $S = [n - i, n + i]$, there are multiple pairs (k, k') such that k is closer to n than k' (provided $i > 0$).[2]

It is then possible to average each side of inequality (6) over all possible such pairs. This is done in (7), which we call the aggregate Ratio Inequality, used to test the theory.

(7) $$\frac{1}{|K|} \sum\nolimits_{(k,k') \in K} \frac{\mathbb{P}[k \mid \textit{around n}]}{\mathbb{P}[k' \mid \textit{around n}]} \geq \frac{1}{|K|} \sum\nolimits_{(k,k') \in K} \frac{\mathbb{P}[k \mid \textit{between m and m}']}{\mathbb{P}[k' \mid \textit{between m and m}']}$$

where $K = \{(k, k') \in S \ ; \ |n - k| < |n - k'|\}$.

3 Experimental paradigm

In this Section, we spell out the general paradigm used in the three Experiments presented in Section 4. This paradigm involves two main tasks for each expression: an Interval task and a Histogram task. The dependent measure extracted from the latter task, consists in a slight adaptation of the peakedness ratios introduced in (7). All three experiments will feature the same tasks and dependent variable. They will only differ in the arrangement of the critical items, and the presence or absence of fillers.

3.1 Tasks

In order to collect empirical posterior distributions corresponding to $\mathbb{P}[k \mid \textit{around n}]$ and $\mathbb{P}[k \mid \textit{between m and m}']$, and test (7)'s validity, we designed a web-based posterior-elicitation experiment. Participants were presented with a short written context, followed by a critical statement containing an expression of the form *around n* or *between m and m′*. Each participant was tested on one critical pair of expressions of the form *around n*/*between m and m′*. The main challenge, from an empirical perspective, was to ensure that the values n, m, and m' were such that the *around n* and the *between m and m′* expressions presented to each participant verified the precondition that their posteriors share the same support.

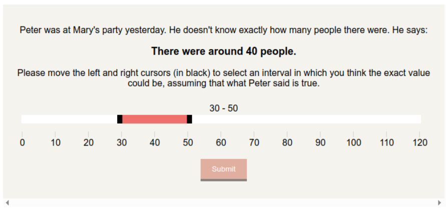
Values of n were randomly set to 40, 50 or 60 for each participant. Values of m and m' were dynamically determined for each specific participant, in a way that will be made clear in the next paragraphs. This was meant to ensure each participant was exposed to *around* and *between* expressions whose induced posteriors shared the same support. The template used for the context and critical sentence is given in Figure 2.

> Peter was at Mary's party yesterday. He doesn't know exactly how many people there were. He says:
> **There were {around n | between m and m′} people.**

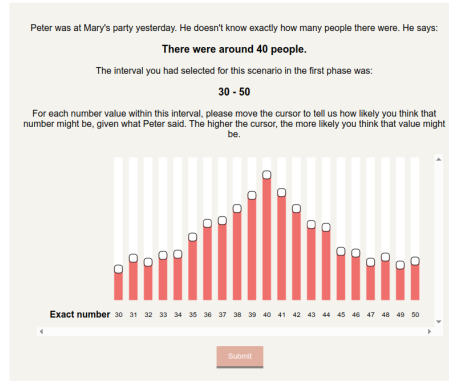
Figure 2: Template for the target sentence and its context.

A trial for a fixed *around n* or *between m and m′* expression was comprised of two main tasks: an Interval task (see Figures 3a and 4a) and a Histogram task (see Figures 3b and 4b). For each expression, the Interval task would always precede the Histogram task, and moreover the Interval task related to the *around* expression would always precede the one related to the *between* expression.

[2]As a function of $i \geq 0$, this number of pairs $f(i)$ is equal to $2 \times (\binom{i+1}{2} + \binom{i}{2}) = 2i^2$.

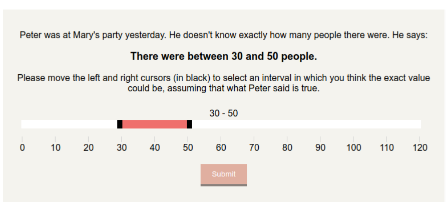


(a) Interval task

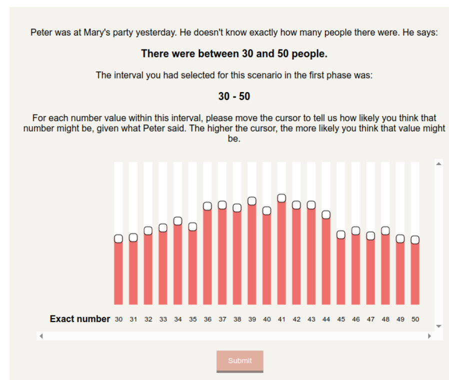


(b) Histogram task

Figure 3: Screen captures of the two tasks for the target expression *around 40* (40 picked randomly in the set $\{40, 50, 60\}$). The values that appear here were defined by the authors and not a participant.



(a) Interval task



(b) Histogram task

Figure 4: Screen captures of the two tasks for the target expression *between 30 and 50*, (30 and 50 determined based on the interval selected in Figure 3a). The values that appear here were defined by the authors and not a participant.

The goal of the Interval task was twofold. First, it was used to elicit the support S related to the posterior of a given expression (*around* or *between*), according to each participant. In practice, participants were asked to move two sliders on a horizontal bar representing the $[0, 120]$ interval (see Figures 3a and 4a) to set the value of S. The two bounds were supposed to define the interval within which the exact numerical value should be, given the target sentence and its context. The second goal of this task was to determine the values of m and m' to use in the *between* expression corresponding to each *around* expression, in order to obtain two expressions comparable by the means of inequality (7). To achieve this, the values of the lower and upper bounds returned by the participant during the Interval task testing *around* were reused as respectively the m and m' values of the critical *between m and m'* expression. For instance, if *around 40* gave rise to a support $S = [27, 52]$ according to a certain participant, the corresponding *between* expression presented to them would be *between 27 and 52*. This was meant to guarantee that the support of the posterior distributions induced by *around n* and *between m and m'* were the same, *for each individual participant*.

The Histogram task was designed to elicit the posterior distributions $\mathbb{P}[k \mid \textit{around } n]$ and $\mathbb{P}[k \mid \textit{between } m \textit{ and } m']$, defined on the supports returned by the participant during the Interval tasks. During this task, participants were again presented with the critical sentence and its context, and were reminded of the interval (support) returned in the preceding Interval task. They were then asked to "draw" a posterior distribution as a histogram, by adjusting one vertical slider per possible number within the support (see

Figures 3b and 4b – drawn by the authors). The resulting histogram was intended to reflect the likelihood of each possible value within the support, given the critical approximation statement.

3.2 Dependent variable and prediction tested

In the experiments described in the next Section, a slight variant of the aggregate Ratio Inequality (7) was tested. First, the pairs (k, k') such that either k or k' is a "salient" value – e.g. "40" in "*around 40*", "30", and "50" in "*between* 30 *and* 50" – were excluded from the set of pairs over which individual posterior ratios were averaged (cf. inequality (8)). Those values were excluded because their very salience might affect the behavior of participants in an experimental setting, leading them to raise their weights compared to others, creating a confounding effect.[3]

$$\text{(8)}\quad \frac{1}{|K|}\sum_{\substack{(k,k')\,\in\,K\\ k,k'\notin\{n,m,m'\}}}\frac{\mathbb{P}[x{=}k \mid x\textit{ is around } n]}{\mathbb{P}[x{=}k' \mid x\textit{ is around } n]} \geq \frac{1}{|K|}\sum_{\substack{(k,k')\,\in\,K\\ k,k'\notin\{n,m,m'\}}}\frac{\mathbb{P}[x{=}k \mid x\textit{ is between } m\textit{ and } m']}{\mathbb{P}[x{=}k' \mid x\textit{ is between } m\textit{ and } m']}$$

Additionally, differences in the supports S_a and S_b of the posteriors induced by respectively *around* and *between* – which were meant to be minimized by the paradigm presented in the previous Section – were accommodated by only considering pairs (k, k') in their intersection $(S_a \cap S_b)$. This yields inequality (9), where K is replaced by $K' = \{(k, k') \in S_a \cap S_b\ ;\ |n - k| < |n - k'|\}$.

$$\text{(9)}\quad \underbrace{\frac{1}{|K'|}\sum_{\substack{(k,k')\,\in\,K'\\ k,k'\notin\{n,m,m'\}}}\frac{\mathbb{P}[x = k \mid x\textit{ is around } n]}{\mathbb{P}[x = k' \mid x\textit{ is around } n]}}_{R_a} \geq \underbrace{\frac{1}{|K'|}\sum_{\substack{(k,k')\,\in\,K'\\ k,k'\notin\{n,m,m'\}}}\frac{\mathbb{P}[x = k \mid x\textit{ is between } m\textit{ and } m']}{\mathbb{P}[x = k' \mid x\textit{ is between } m\textit{ and } m']}}_{R_b}$$

We call R_a (for "*around*-ratio") and R_b (for "*between*-ratio") the left-hand side and right-hand side of inequality (9), respectively. Inequality (9) is the one we propose to test in the series of experiments presented in the next Section.

4 Experiments

4.1 Experiment 1

The first experiment was conducted on 207 participants (determined *via* a power analysis based on pilot data), and preregistered on OSF. Participants were US-based Amazon Mechanical Turk Master Workers with a success rate of at least 98% on previous MTurk tasks. Participants were excluded based on (preregistered) demographic factors and quality requirements. First, participants who signaled that they were not native speakers of English were excluded. The remaining participants who were 1.645 standard deviations faster than the average completion time were also excluded. Finally, participants who provided inconsistent intervals on critical items were excluded. Inconsistency was defined as non-overlap with $[m, m']$ in the case of *between m and m'*, and as non-overlap with the central value n in the case of *around n*. These conditions were relatively permissive, which meant that *around*/*between* pairs not sharing the exact same support were included.

Each trial was comprised of an Interval task immediately followed by the corresponding Histogram task. In addition to the two critical trials, this experiment involved six filler expressions featuring *around*, *between*, and *almost*, combined with numbers of varying granularities. The target *around* and *between* items were presented at two fixed positions in the experiment, as 4th and 8th trial respectively. The positions of the filler items were randomized over the set of remaining positions. The choice to associate filler expression

[3]Furthermore, "around 40" might trigger, through competition with the unmodified form "40", the inference that 40 is unlikely to be the exact value, because if it were, a fully informed speaker would use "40". Such an effect would go against our prediction and is not captured in our model. A fuller model would include additional pragmatic inferences of this sort that we do not address here.

with different levels of granularity was intended to conceal the link between the critical *around n* expression (position 4) and the critical *between m and m′* expression (position 8). Table 1 summarizes the design of this experiment.

Expression	Position	Category
around n	4	Critical
between m and m′	8	Critical
between 80 and 90	random($\{1, 2, 3, 5, 6, 7\}$)	Filler
between 13 and 24	random($\{1, 2, 3, 5, 6, 7\}$)	Filler
around 70	random($\{1, 2, 3, 5, 6, 7\}$)	Filler
around 36	random($\{1, 2, 3, 5, 6, 7\}$)	Filler
almost 60	random($\{1, 2, 3, 5, 6, 7\}$)	Filler
almost 24	random($\{1, 2, 3, 5, 6, 7\}$)	Filler

Table 1: Trials for Experiment 1

Because the Interval and Histogram tasks related to a fixed expression were set to be adjacent within a trial, and because of the dependency between the m and m' values of the critical *between m and m′* expression and the critical trial testing *around n*, the order of these two trials could not be randomized.

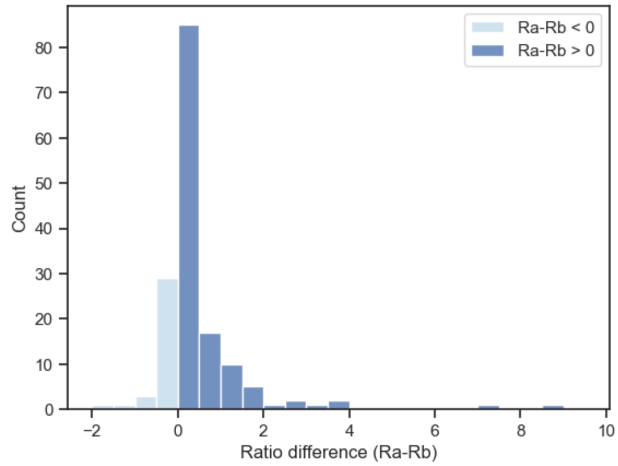
A two-tailed Sign test for matched pairs[4] with a significance threshold of $.05$ was used to test the hypothesis $R_a \neq R_b$ within-participant, on a sample size of 162 (after all exclusions). It turned out significant in favor of $R_a > R_b$ ($p = 6.6$e-13), in line with inequality (9). 126 participants (78% of the sample) were associated with a positive ratio difference (in line with the hypothesis), while 34 participants (21% of the sample) were associated with a negative ratio. Two participants were linked to equal ratios.

Exploratory two-tailed Wilcoxon tests performed on each subset of participants who were assigned to the same value of n, turned out significant under Holm-Bonferroni correction ($p = 1.4$e-$4 < 1$e-3 for $n = 40$; $p = 7.7\text{e-}7 < \frac{1\text{e-}5}{3}$ for $n = 50$; $p = 1.0\text{e-}4 < \frac{1\text{e-}3}{2}$ for $n = 60$). Figure 5a illustrates the distribution of paired ratio differences.

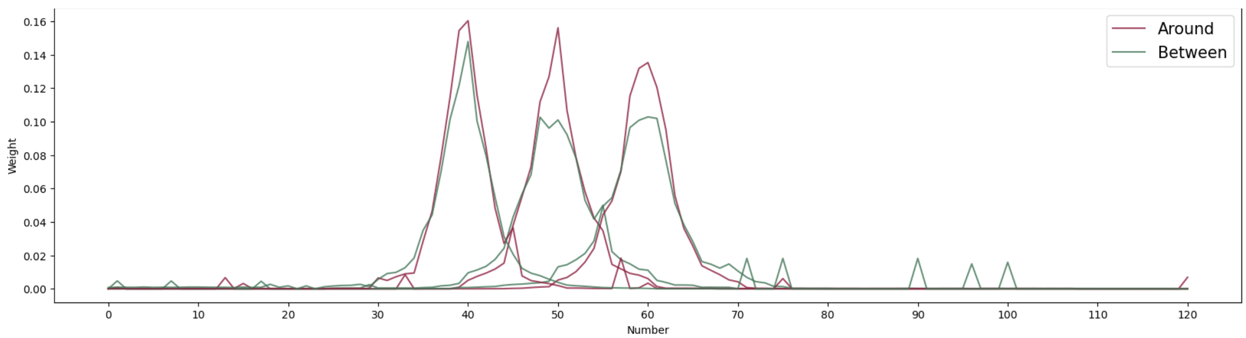
Group-by-group averaged results are given in Figure 5b and confirm the within-participant results, in that posteriors induced by *around* tend to be more peaked than posteriors induced by a similar *between* expression.[5]

[4]The preregistration originally planned a two-tailed Wilcoxon signed-rank test, which turned out significant in favor of $R_a > R_b$ ($p = 8.5$e-13). We chose to report a Sign test instead (despite this test having less power), because we realized that a Wilcoxon test, being sensitive to the magnitude of individual ratio differences, may turn out significant globally by the effect of only one number group (e.g., $n = 60$) if this group is associated with high ratio differences in the expected direction.

[5] When averaging posteriors across participants, we expect to get a peaked distribution even if each participant had reported a completely uniform distribution on the interval they had chosen. Imagine indeed that all participants report a uniform distribution, but choose intervals of different lengths. Since all intervals contain the target number (as a result of our exclusion criteria), numbers that are closer to the target will belong to more intervals than those that are further away. Therefore, when we average over participants, we expect that the closer a number is to the target, the more average weight it will get. For this reason, the 'peakedness' of the curves shown on Figure 5b cannot be interpreted as the average peakedness over participants, as it is likely to be more pronounced than in individual distributions. However, the visual contrast between around and between is still meaningful, since, given our design, each participant is expected to choose the same interval for *between* and *around*. The curves that result from restricting the data to the participants who chose exactly the same intervals for the two sentence types are visually similar to those we show here and this is so also for comparable graphs in relation to subsequent experiments.



(a) Distribution of paired ratio differences ($R_a - R_b$).



(b) Averaged posteriors for $n = 40, 50$, or 60.

Figure 5: Results of Experiment 1.

However, we also observed that participants tended to provide distributions that were more uniform as they advanced through the study. This constitutes a potential confound that we went on to address in two follow-up experiments (Experiment 2 and 3).

4.2 Experiment 2

Experiment 2 was preregistered on OSF, and conducted on 150 participants.[6] Inclusion and exclusion criteria were the same as Experiment 1.

In order to decrease the distance between the two critical trials and in turn reduce the "flattening" effect driven by the number of completed trials, fillers were removed. The design of this experiment is given in Table 2. Just like in Experiment 1, Interval and Histogram tasks related to the same expression were set to be adjacent.

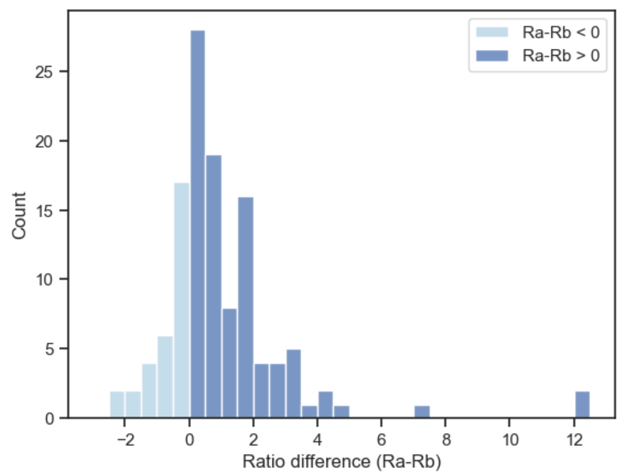
Expression	Position	Category
around n (Interval+Histogram)	1	Critical
between m and m′ (Interval+Histogram)	2	Critical

Table 2: Trials for Experiment 2

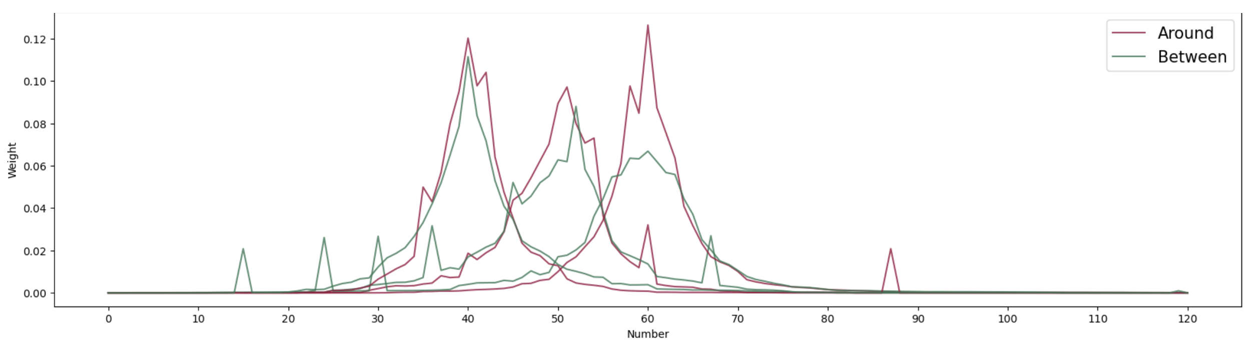
[6]We ajusted the number based on the exclusion rate observed in Exp. 1

A preregistered Sign test (one-tailed) with a significance threshold of .05, applied to a sample of size 122 after all exclusions, turned out significant in favor of $R_a > R_b$ ($p = 2.5$e-8), again in line with inequality (9). 91 participants (74.5% of the sample) were associated with a positive ratio difference, while 31 participants (25.5% of the sample) were associated with a negative difference. Figure 6a illustrates the distribution of paired ratio differences.

Preregistered, number-specific Wilcoxon tests (one-tailed) ended up significant as well under Holm-Bonferroni correction ($p = 4.3\text{e-}4 < \frac{1\text{e-}3}{2}$ for $n = 40$; $p = 6.5\text{e-}4 < 1\text{e-}3$ for $n = 50$; $p = 8.5\text{e-}6 < \frac{1\text{e-}4}{3}$ for $n = 60$). By-number averaged results are displayed in Figure 6b and confirm the within-participant results.



(a) Distribution of paired ratio differences ($R_a - R_b$).



(b) Averaged posteriors for $n = 40$, 50, or 60.

Figure 6: Results of Experiment 2.

Even though Experiment 2 corroborated the result of Experiment 1, its design did not allow us to totally remove the potential order effect between *around* and *between*, since the Histogram task related to *around* still systematically preceded the Histogram task related to *between*. This issue is what motivated Experiment 3.

4.3 Experiment 3

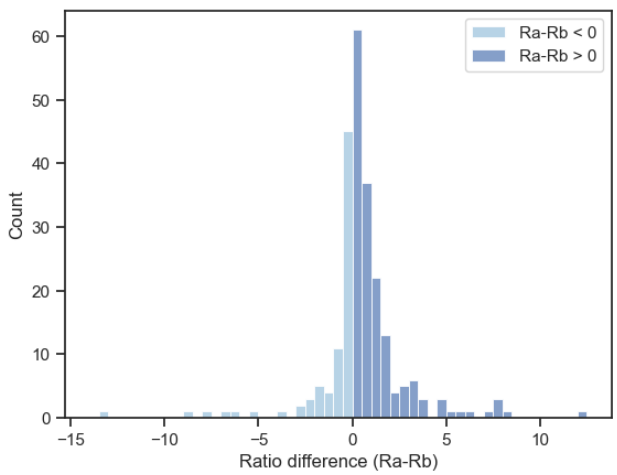
Experiment 3 was preregistered on OSF and conducted on 286 participants. The sample size was determined based on a power analysis using as target effect size half the effect-size of Experiment 2 (which led to a target sample size of 250), and based on the exclusion rates from Experiments 1 and 2. Inclusion and exclusion criteria were the same as for Experiments 1 and 2.

Experiment 3 differs from Experiment 2 only in that the two Histogram tasks are fully randomized across participants. This was achieved by decoupling these tasks from their corresponding INTERVAL tasks.[7] Specifically, Experiment 3 started with two (non-randomized) INTERVAL tasks, testing first *around n*, and then *between m and m′* (m and m' still being dynamically determined). The two INTERVAL tasks were then followed by the corresponding HISTOGRAM tasks, in a randomized order. Table 3 summarizes the design of this experiment.

Expression	Position	Category
around n (INTERVAL)	1	Critical
between m and m′ (INTERVAL)	2	Critical
around n (HISTOGRAM)	random($\{3, 4\}$)	Critical
between m and m′ (HISTOGRAM)	random($\{3, 4\}$)	Critical

Table 3: Trials for Experiment 3

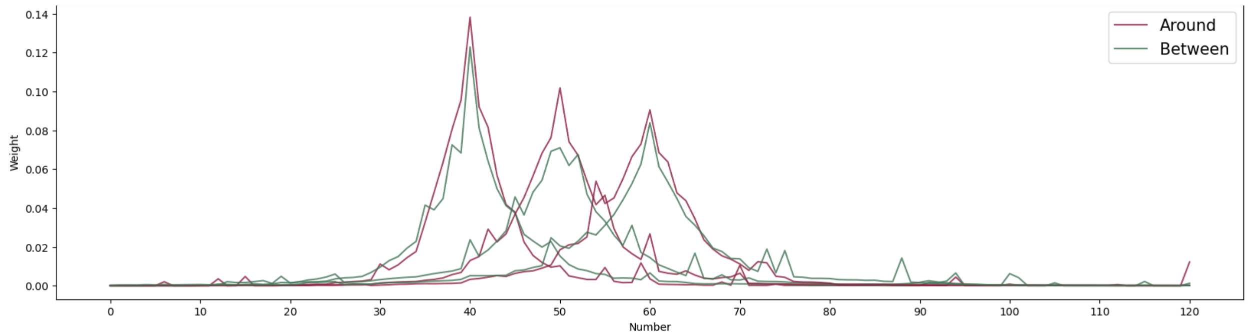
A preregistered one-tailed Sign test performed on a sample of 240 participants after all exclusions, turned out significant in favor of $R_a > R_b$ ($p = 1.5$e-8). 163 participants (68% of the sample) were associated with a positive ratio difference, while 77 participants (32% of the sample) were associated with a difference going in the opposite direction. Figure 7a illustrates the distribution of paired ratio differences.[8] Group-by-group averaged results are given in Figure 7b.



(a) Distribution of paired ratio differences ($R_a - R_b$).

[7]We thank REDACTED for suggesting this modification.

[8]In this Experiment, we chose not to preregister additional number-specific one-tailed Wilcoxon tests. This is because power calculations suggested that the target effect size for such tests – which was expected to be reduced by the neutralization of order effects – could only be reached with significantly bigger sample sizes (> 450, given past exclusion rates). For comparison with Experiments 1 and 2, we performed these tests in an exploratory way. They ended up significant under Holm-Bonferroni corrections ($p = 1.0$e-5 $< \frac{1\text{e-}4}{3}$ for $n = 40$; $p = 3.7$e-4 $<$ 1e-3 for $n = 50$; $p = 2.1$e-4 $< \frac{1\text{e-}3}{2}$ for $n = 60$).



(b) Averaged posteriors for $n = 40$, 50, or 60.

Figure 7: Results of Experiment 3.

In sum, the fully randomized design of Experiment 3 confirms the results of both previous experiments.

5 Discussion

Our studies used a within-participant design to test and confirm the peakedness contrast predicted by Égré et al., 2023 for pairs of *around*- and *between*-expressions. However, even though *around*- and *between*-expressions were dynamically presented to each individual participant in order to be maximally similar in terms of the support of their induced posteriors, we anticipated that this condition would not be met by all participants tested, which is why form (9) of the Ratio Inequality was used. To assess the effect of this restriction, we conducted post hoc analyses looking more closely at the properties of the *intervals* chosen by our participants during the Interval task and of the *supports* of the corresponding posteriors reported in the Histogram task.[9]

The proportion of participants with strictly matching intervals/supports for critical *around* and *between* trials varies between about a half to a third (Experiment 1: 53.1% with matching intervals; 50.6% with matching supports; Experiment 2: $36.1\%/32.0\%$; Experiment 3: $34.2\%/29.6\%$.) The remaining cases are split in two groups: a group in which the interval/support corresponding to *around* was *containing* the interval/support corresponding to the matching *between*-expression; and a group in which the interval/support corresponding to *around* was *contained* in the interval/support corresponding to the matching *between*-expression. The latter group appears to be the most problematic *a priori*, because for this group, the *around*-posterior, being defined on a comparatively smaller interval, may mechanically be more peaked than the corresponding *between*-posterior, *independently* of our original prediction.[10]

Approximately 50% of the participants belonged to this potentially problematic group across experiments, with little variation depending on whether intervals or supports were considered. However, post-hoc linear mixed-effect regressions indicate that, across experiments, sentence type (*around* vs *between*) is a significant predictor of the peakedness ratio, even controlling for the potential effect of the length of the interval or support.[11] This was done by fitting a LMER of the form $log(R) \sim n + a + l + (1|i)$, with R the ratio variable, n the target number, a the approximator (*between* vs. *around*), l the length of the interval or the length of the support of the posterior, and i the participant (we therefore fitted 6 models in total, 2 for each experiment, depending on whether we used the support or the interval of the posterior as a predictor). Across experiments, comparisons between these models and simpler models without a as a factor always gave rise to a significant p-value, even after correction for multiple comparisons. This suggests that the effect we found is not specifically driven by the subset of the participants who associated *around* with a smaller interval/support than *between*.

[9] Supports are the sets $[k, k']$ such that k (resp. k') is the smallest (resp. highest) element of the range of the participant's posterior that is assigned a non-zero probability.

[10] For instance, if participants had a tendency, irrespective of the sentence type, to assign the maximal weight to the target, minimal weights to the bounds, with a linear decrease from the target to the bounds, shorter intervals would necessarily give rise to higher peakedness, corresponding to a greater ratio score

[11] See the OSF related material for details of these analyses.

Lastly, restricting our initial analyses of the ratio differences to the uncontroversially unproblematic participants (i.e., those with matching intervals/supports for the two sentence types) consistently gives rise to significant p-values by a sign-test using Holm-Bonferroni correction for multiple comparisons.[12] Overall, these analyses suggest that the slight variations observed in the intervals or supports of the relevant posteriors do not represent confounding factors when it comes to our assessment of the Ratio Inequality.

6 Conclusion

The model developed by Égré et al., 2023 proposes that the interpretation of an *around*-statement in a given context is inherently vague, unlike that of a *between*-statement. This is modeled by assuming that when a listener processes *around n*, the interval semantically denoted by "around" is open: it is treated as a random variable in the truth conditions of "around", and it drives a Bayesian update which differs from that for "between" by depending on the various possible values that this random variable can take.

Our results support the theory presented in Égré et al., 2023, and thereby provide a non-trivial confirmation of the Bayesian approach to linguistic interpretation of vague utterances more generally. They show us that while *x is around n* and *x is between m and m* both convey factual uncertainty about the exact location of x (when m and m' are distinct), for *around* this factual uncertainty interacts with the semantic uncertainty inherent to the meaning of *around* to convey distinctive probabilistic information, not conveyed by any matching *between*-statement.

One limitation of this study regarding the phenomenon of interest is that we have only looked at how *around* and *between* are interpreted, not at how they are produced. The study does not tell us in which cases a speaker would prefer to use "between" or to use "around". The theory presented in Égré et al., 2023 considers this problem, and predicts that an agent might prefer to communicate their uncertain numerical estimate of a quantity using "around" when the speaker's estimate is itself biased toward a specific value, and also when they want to convey some information about the underlying shape of their probability distribution. We leave to further work an experimental investigation of the production of distinct approximation expressions and their competition in different contexts.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work the author(s) did not use any generative AI tool as part of the writing process. The author(s) take full responsibility for the content of the published article.

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[12] Experiment 1: 55/82 participants with matching supports were associated with ratio differences in line with the hypothesis; 59/86 if intervals were considered instead. Experiment 2: 30/39 (with supports) / 32/44 (with intervals). Experiment 3: 47/71 (with supports) / 55/82 (with intervals).

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